DIFFERENTIABLE MANIFOLDS

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These are my notes for Topology II taught by Eduard Looijenga at UChicago in Winter 2026.

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1. Basic Definitions

A *topological manifold* of dimension m is a topological space M that is Hausdorff and locally homeomorphic to $\mathbb{R}^m$. Such an M has an open covering $\mathcal{A} = \{U_\alpha\}$ called an *atlas* with associated homeomorphisms (*charts*) $\kappa_\alpha : U_\alpha \to \mathbb{R}^m$ which are compatible, meaning that in each intersection $U_\alpha \cap U_\beta$, we have a homeomorphic coordinate change map:

$$\mathbb{R}^m \supset \kappa_\alpha(U_\alpha \cap U_\beta) \xrightarrow{\kappa_\beta \kappa_\alpha^{-1}} \kappa_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^m$$

The atlas is C^k if all the coordinate change maps are C^k.

$\mathcal{A}$ must be C^k in order to define the notion of a C^k function $M \to \mathbb{R}$ (relative to $\mathcal{A}$); otherwise, we could have $f : M \to \mathbb{R}$ that is C^k through one chart but not another. Naturally, which functions $M \to \mathbb{R}$ are C^k depends on $\mathcal{A}$. And in fact, **atlases define the same notion of C^k iff they are compatible**. That is, all possible notions of a C^k function on M correspond to maximal atlases, or "*C^k structures*."

The presence of a C^k structure enriches M and allows one to say more about it, so it is natural to ask whether a given M has a C^k structure. Whitney showed that all manifolds with a C^k structure also have a C^∞ structure that can be obtained by restricting the corresponding atlas (and hence a C^j structure for $j > 0$). So the C^k structures come together. However, there are topological manifolds with no C^1 structure, and hence no C^k structure for any $k > 0$. Thus the only distinction is between smooth manifolds and non-differentiable manifolds. We will be concerned only with the former.

A bijection $f : M \to N$ between smooth manifolds is called a *diffeomorphism* if it and its inverse are both C^1. This is more strict than a *homeomorphism*, which is only required to be continuous in both directions.

While in Algebraic topology we are concerned with the homotopy types of spaces, which is a coarser characterization than homeomorphism type, when studying smooth manifolds we can also ask about the diffeomorphism type, which is finer[1].

1.1. Submanifolds, Immersions, Submersions, and Embeddings.

All m-manifolds M, because they are locally homeomorphic to $\mathbb{R}^m$, have an associated *tangent space* T_pM at each point p. When M is inside a Euclidean space, this is literally the set of tangent vectors to M at p. In general it can be defined for all manifolds M in a compatible way. T_pM is in fact a vector space of dimension m. For smooth maps $f : M \to N$, we can think of the derivative D_pf as literally a linear map between tangent spaces $D_pf : T_pM \to T_{f(p)}N$.

The quality of D_pf locally tells us a lot about its overall properties.

- f is called an *immersion* if D_pf is injective at all points p.
- f is called a *submersion* if D_pf is surjective (i.e. full rank) at all points p.
- f is called a *local diffeomorphism* if D_pf is invertible at all points p.

Note that a local diffeomorphism need not be a diffeomorphism because though it is locally invertible it might not be globally (consider e.g. the map $\mathbb{R} \to S^1$ given by $x \mapsto e^{ix}$).

The image of a local diffeomorphism need not even be a manifold. Consider the map taking S^1 to a figure-eight. Locally, every section of the circle is sent to a segment of the figure-eight, yet near the crossing point there is no homeomorphism to $\mathbb{R}$, so the figure-eight isn't a manifold. Another way of looking at this is that the tangent space to the figure-eight at the crossing point is *not* a vector-space. For a higher-dimensional example, take the Klein bottle. There is a local diffeomorphism to a "fake Klein bottle" in $\mathbb{R}^3$ which self-intersects in a circle.

If an immersion is injective *and* homeomorphic onto its image, it is called an *embedding*[2]. The image of an embedding $f : M \to N$ is always a manifold, and is called a *submanifold* of N. Submanifolds can be equivalently characterized in another way: a manifold $A \subset N$ is a submanifold

[1] For example, the homeomorphism type of S^7 splits into 28 diffeomorphism types, as shown by Milnor.

[2] The homeomorphism part is necessary because for example one can construct an injective immersion $(0, 1) \to \mathbb{R}^2$ whose image is a figure-eight.

of N if the charts $\kappa : N \to \mathbb{R}^n$ send A to a linear subspace $\mathbb{R}^k \subset \mathbb{R}^n$.

Question: Can a given N be realized as a submanifold of a given M (especially when $M = \mathbb{R}^m$)? Or in another way, can we classify all submanifolds of a given M up to diffeomorphism type?

- Any smooth manifold (as long as its topology has a countable basis) can be embedded in $\mathbb{R}^m$ for sufficiently large m, though exactly what m is the minimum is not always easy to determine.
  - **Whitney's Embedding Theorem**: a smooth m-manifold can always be embedded in $\mathbb{R}^{2m}$ and immersed in $\mathbb{R}^{2m-1}$ (smaller dimensions may be possible).
  - Cohen proved more generally that a smooth m-manifold could be immersed in $\mathbb{R}^{2m-a(m)}$, where $a(m)$ is the number of 1's in the binary expansion of m.
  - For example, the Klein bottle K can be described without reference to an underlying space via gluing instructions, and the lowest-dimensional space it can be embedded in is $\mathbb{R}^4$. In $\mathbb{R}^3$ it can be *immersed*, but not embedded.
- **Inverse/Implicit Function Theorem**:
  - If $f : M \to N$ is an immersion at p, $f(U)$ is a submanifold of N for some open $U \ni p$.
  - If $f : M \to N$ is a submersion at $p \mapsto q$, then the fiber $f^{-1}(q)$ is a submanifold of M.
- **Transverse Intersections**:
  - We say $f : M \to N$ is *transverse* to a submanifold $Q \subseteq N$ if for $q = f(p) \in Q$, $D_q f(T_p(M))$ and $T_q(Q)$ span $T_q(N)$. This is denoted $f \pitchfork Q$.
  - In this case, $f^{-1}(Q)$ is a submanifold of M with the same codimension as Q in N.

1.2. **Tangent Bundles and Vector Fields.** Every smooth manifold M has an associated *tangent bundle* TM whose elements are pairs (p, v) where $v \in T_pM$. It is a $2m$-dimensional manifold.

A *vector field* over a manifold is a smooth map $V : M \to TM$ with $V(p) \in T_p(M)$. For some manifolds M, it is possible to give a basis for TM by vector fields; that is, to give vector fields $V_1, V_2, \ldots, V_m$ such that their values at $p \in M$ are always a basis of T_pM. Such manifolds are called *parallelizable*.

Question: Which M are parallelizable?

- Examples:
  - $M = S^1$ is parallelizable. A basis can be given by a 90-degree rotation at every p. Thinking of S^1 as $\mathbb{C}^\times$, this corresponds algebraically to a multiplication by i.
  - $M = S^2$ is not. In fact, there isn't even a nonzero vector field $V : M \to TM$. Suppose there were such $V : M \to TM$. Such V induces by projection a map $V' : S^2 \to S^2$ with $V'(p) \perp p$ for all $p \in S^2$. But we know from Algebraic Topology that every map $S^n \to S^n$ either maps at least a point to its antipode or has a fixed point (if $p, V'(p)$ are not antipodes then there is a unique shortest path between them, so we can homotope every $V'(p)$ continuously along this shortest path to p, giving a homotopy to the identity map, and hence there is a fixed point).
  - $M = S^3$ is parallelizable. It is given by analogy to the Quaternions: thinking of S^3 as $\mathbb{H}^\times$, then we have a basis via the three vector fields $p \mapsto ip, p \mapsto jp, p \mapsto kp$.
  - As it turns out, S^1, S^3, S^7 are the only spheres that are parallelizable.
- In general, all Lie groups are parallelizable (we show this on HW).

A *derivation* is an $\mathbb{R}$-linear map $D : \mathcal{E}(U) \to \mathcal{E}(U)$ (where $\mathcal{E}(U)$ is the space of functions on U) which obeys the Leibniz rule $D(fg) = Df \cdot g + f \cdot Dg$. The derivative along any vector is a natural example of such a map, and more generally a vector field can act on $\mathcal{E}(U)$ in the same way. In fact, **all derivations correspond to actions of vector fields**; that is, every derivation is of the form

$$V(f)(p) = (f \circ \gamma)'(0) = D_p(f) \cdot V(p)$$

where $\gamma : \mathbb{R} \to U$ is a smooth path such that $\gamma(0) = p$ and $\gamma'(p) = V(p)$ (note that this only depends on the vector field V).

To go the other way, the vector field associated with derivation D is

$$V(p) = (D(\gamma_1)(p), D(\gamma_2)(p), \ldots, D(\gamma_m)(p))$$

where $\gamma_j : \mathbb{R} \to M$ are paths with $\gamma_j(0) = p$ which form a basis of the tangent space T_pM.

For any derivations V, W on M, the *Lie bracket* $[V, W] := V \circ W - W \circ V$ is a derivation (simple exercise), and hence the Lie bracket can be defined on the space of smooth vector fields on M, making this space into a Lie Algebra. The Lie bracket is bilinear, anti-symmetric, and satisfies the Jacobi identity:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

1.3. **Flows.**

- A *flow* is a smooth map $H : \mathbb{R} \times M \to M$ with the properties
  - $H_0 = \mathrm{id}_M$
  - $H_t H_s = H_{t+s}$.
- We write $H_t : M \to M$ for $H_t : p \mapsto (t, p)$. The map $t \mapsto H_t$ is a group homomorphism $\mathbb{R}$ to the group of diffeomorphisms on M.
- The *orbit through* p, denoted γ_p, is $\gamma_p : t \mapsto H_t(p)$.
- For each H we have an associated vector field $V_H : p \mapsto \gamma_p'(0) \in T_pM$. This is called the *infinitesimal generator* of H.
  - This correspondence is not bijective; there are vector fields that do not come from flows.
  - However, there is a bijective correspondence between *maximal local flows* and vector fields. A local flow is like a flow except it need not be defined on all of $\mathbb{R} \times M$, rather for each $m \in M$ it is defined on some open interval around 0. It is maximal if this domain $\Omega \subset \mathbb{R} \times M$ is as large as possible, so on the boundary H should be diverging.
  - The maximal local flow corresponding to V is called the *evolution* of V.
- Question: which V generate a flow (as opposed to a local flow)?
  - If the evolution of V contains a fixed time interval $(-\varepsilon, \varepsilon)$ at every point of M, then it generates a flow (HW).
  - If V is compactly supported, then it generates a flow (HW).

One natural application of this concept is to describe the vector field $[V, W]$: if V generates a flow H, then for any vector field W,

$$\partial_t|_{t=0} H_t^* W = [V, W]$$

where $H_t^* W$ is the vector field W pulled back for time t along H, i.e. $[V, W]$ is the direction and magnitude along which W moves as you follow the flow of V.

In another way, if V, W generate flows H, I, then we can write $[V, W]$ as a derivative in terms of these flows:

$$\partial_s \partial_t \big|_{s=t=0} I_t H_s I_t^{-1} H_s^{-1} = [V, W].$$

Intuitively, $[V, W]$ measures the failure of H and I to commute. Conversely, if $[V, W] = 0$, then $H_s I_t = I_t H_s$ (HW).

1.4. **Vector Bundles.** A *vector bundle* over M is type of projection map $\pi : E \to M$ from some "total space" E such that the fibers $E_p := \pi^{-1}(p)$ have the structure of r-dimensional vector spaces. This vector space identification must be smooth in p, meaning precisely that M can be covered by open sets U such that E_U is *locally trivial* via $\rho : E_U \to \mathbb{R}^r$, which satisfies the following properties:

- Restricted to a single fiber, $\rho : E_p \to \mathbb{R}^r$ is a vector space isomorphism.
- E_U is diffeomorphic to $U \times \mathbb{R}^r$ via $(\pi, \rho) : E_U \to U \times \mathbb{R}^r$.

Examples:

- The trivial bundle $\mathbb{R}^r \times M$ with $E = \mathbb{R}^r$ and $\rho : \mathbb{R}^r \to \mathbb{R}^r$ the identity.
- The tangent bundle TM.

- The "tautological line bundle" γ_m^1 over $\mathbb{P}^m$, where $E = \{(x, \ell) : x \in \mathbb{R}^{m+1} \cap \ell, \ell \in \mathbb{P}^m\}$, and π is the projection in the second element.

In each of the intersections $U_\alpha \cap U_\beta$, at a point $p \in U_\alpha \cap U_\beta$ there is a transition map

$$\rho_{\alpha,p} \circ \rho_{\beta,p}^{-1} : \mathbb{R}^r \to E_{U \cap U'} \to \mathbb{R}^r$$

which is an invertible linear map in $\mathrm{GL}_r(\mathbb{R})$. Thus we have a *transition function* $g_{\alpha,\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{R})$ defined by $g_{\alpha,\beta}(p) := \rho_{\alpha,p} \circ \rho_{\beta,p}^{-1}$.

If we want to take a map $E \to M$ with some atlas U_α, ρ_α and make it into a vector bundle, it suffices to know that these maps $g_{\alpha,\beta}$ are smooth. We can use this fact to define duals, tensor products, direct sums, etc. of vector bundles.

Some associated definitions:

- A *sub-bundle* of $E \to M$ is a subspace $E' \subseteq E$ for which $\rho_U(E_U \cap E') = \mathbb{R}^{r'} \times 0$.
- There is a *quotient bundle* E/E' between a bundle E and sub-bundle E'.
- If $M \subset N$ is a submanifold, then TM is a sub-bundle of $TN|_M$. The quotient bundle $\nu_{N,M} := TN|_M/TM$ is called the *normal bundle* of M and N.
- A map $s : M \to E$ such that $\pi(s(m)) = m$ is called a *section* of π. The space of sections is $\mathcal{E}(M)$. Within a chart $\rho : E_U \to U \times \mathbb{R}^r$, a section can be specified by a smooth map $U \to \mathbb{R}^r$.

Using this terminology, we can view vector fields on M as sections of the tangent bundle on M.

2. Differential Forms

In this section we'll concretely define the notion of a differential (like $\mathrm{d}x$ or $x\mathrm{d}x$, etc.) using the language built up so far.

The *cotangent space* of M at p is the dual of the tangent space, T_p^*M. Similarly, the *cotangent bundle* of M is the dual of the tangent bundle, $\tau_M^* : T^*M \to M$.

A *differential* on M is a section of the cotangent bundle, i.e. a map $M \to T^*M$. The basic example is $\mathrm{d}\varphi : p \mapsto \mathrm{d}_p\varphi$ where $\varphi : M \to \mathbb{R}$ is some function. $\mathrm{d}_p\varphi$ acts on a vector field $V : M \to TM$ by $\mathrm{d}_p(\varphi)(V) = V(\varphi)$. Locally (i.e. in some U treated as a subset of $\mathbb{R}^m$),

$$\mathrm{d}f = \sum_{i=1}^{m} \frac{\partial f}{\partial x_j} \mathrm{d}x_j\big|_U.$$

This d is a linear map from functions on U to sections of $\tau_M^*|_U$:

$$\mathrm{d} : \mathcal{E}(U) \to \mathcal{E}(\tau_M^*|_U) =: \mathcal{E}^1(U).$$

A *differential k-form* is a section of $\wedge^k \tau_M^*$. So the differential defined previously is a 1-form. We can think of a k-form on M as a thing which can (skew-linearly) extract a real value at each point of M from a k-tuple of m-dimensional vector fields.

The space of k-forms on a given chart $M \supset U \to \mathbb{R}^m$ is denoted $\mathcal{E}^k(U)$, and we have the exterior derivative

$$\mathrm{d} : \mathcal{E}^k(U) \to \mathcal{E}^{k+1}(U)$$

for all $k \geq 0$. This d in general has some properties:

- $\mathbb{R}$-linearity.
- *Flatness*: $\mathrm{dd}(\alpha) = 0$ for all α.
- *Leibniz Rule*: $\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \mathrm{d}\beta$.
- *Pullback-Commuting*: $\mathrm{d}f^*(\alpha) = f^*(\mathrm{d}\alpha)$ for $f : M \to N$ and α a form on N.

The space of all $k \leq m$ forms on M at U, denoted $\mathcal{E}^\bullet(U)$, is a graded-commutative $\mathbb{R}$-algebra with product $\wedge$.

Relative to a vector field V on M, we have an "inner contraction" map

$$\iota_V : \mathcal{E}^k(M) \to \mathcal{E}^{k-1}(M).$$

ι_V is defined by

$$(\iota_V \alpha)(W_1, \ldots, W_{K-1}) = \alpha(V, W_1, \ldots, W_{k-1})$$

where $W_1, \ldots, W_{k-1}$ are other vector fields.

Using both ι_V and d, we define the *Lie Derivative*

$$\mathcal{L}_V = \iota_V \mathrm{d} + \mathrm{d}\iota_V : \mathcal{E}^k(M) \to \mathcal{E}^k(M).$$

This $\mathcal{L}_V$ captures the change in a form as it flows along V; i.e.

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} H_t^* \alpha = \mathcal{L}_V(\alpha).$$

2.1. **De Rham Cohomology.**

The exterior derivative yields a sequence

$$\mathcal{E}^0(M) \xrightarrow{\mathrm{d}} \mathcal{E}^1(M) \xrightarrow{\mathrm{d}} \mathcal{E}^2(M) \to \ldots$$

which has $\mathrm{d}^2 = 0$, so it has a defined homology and cohomology. This cohomology

$$H_{dR}^k(M) := \frac{\ker(\mathrm{d})}{\mathrm{im}(\mathrm{d})}$$

is called the De Rham Cohomology. The Poincaré Lemma says

$$H_{dR}^k(\mathbb{R}^m) = \begin{cases} \mathbb{R} & k = 0 \\ 0 & k > 0 \end{cases}$$

i.e. that every closed $k > 0$ form on $\mathbb{R}^m$ is exact. And in general if $\pi : M \times \mathbb{R} \to M$ is the projection map, then the induced map

$$H^k_{dR}(M) \xrightarrow{\pi^*} H^k_{dR}(M \times \mathbb{R})$$

is an isomorphism. To prove this, we can show that $\mathrm{d}\mathcal{E}^{k+1}(M \times \mathbb{R})$ and $\mathcal{E}^k(M)$ span $\mathcal{E}^k(M \times \mathbb{R})$.

2.2. **Stokes' Theorem.** Let $\alpha \in \mathcal{E}^m(\mathbb{R}^m)$ with compact support, so

$$\alpha = f(x)\mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_m$$

for some compactly supported function $f : \mathbb{R}^m \to \mathbb{R}$. We define the integral of α over $\mathbb{R}^m$ to be

$$\int_{\mathbb{R}^m} \alpha := \int_{\mathbb{R}^m} f(x)\, \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_m.$$

If we change coordinates $(x_1, \ldots, x_m) \to (y_1, \ldots, y_n)$ via some smooth map $h : \mathbb{R}^m \to \mathbb{R}^m$, we might integrate $h^*(\alpha)$, which gives

$$\begin{aligned}\int_{\mathbb{R}^m} h^*(\alpha) &:= \int_{\mathbb{R}^m} f(h(x))\, \mathrm{d}h_1 \wedge \cdots \wedge \mathrm{d}h_m \\ &= \int_{\mathbb{R}^m} f(y) \cdot |\det(h)(x)|\, \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_m.\end{aligned}$$

$\det(h)$ is everywhere nonzero because it is smooth. If in addition it is positive everywhere (which we can assume as long as the support of f is connected), then

$$\int h^*(\alpha) = \int \alpha.$$